OHM-1

POLYPHONIC VIRTUAL ANALOG SYNTHESIZER

OWNER'S MANUAL

WEB AUDIO API · MIDI-READY · IN-BROWSER · NO INSTALLATION REQUIRED

1 SIGNAL FLOW

Understanding the signal flow of OHM-1 is essential for getting the most out of the synthesizer. Audio begins at the **Source Module** and passes through the **Filter**, **Amplifier**, and **Effects Rack** before reaching the output. Modulation sources (LFOs, envelopes) and triggers (MIDI, Sequencer) operate in parallel.

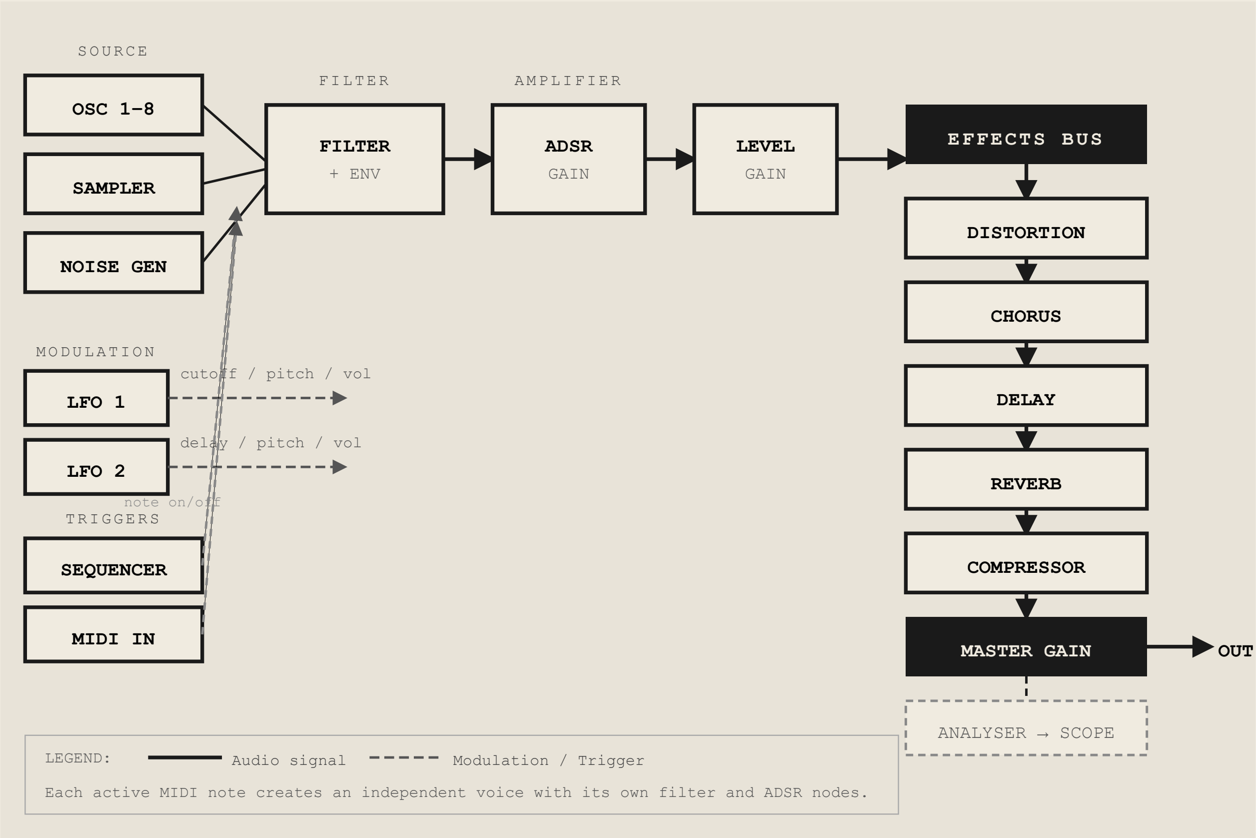

FIG. 1-1 — OHM-1 COMPLETE SIGNAL CHAIN

Each **MIDI note-on** event spawns an independent voice containing its own oscillator bank, filter node, and ADSR gain node. All voices converge at the **Effects Input Bus** and are processed by the shared effects chain before reaching the master output.

2 SOURCE MODULE

2.1 OSCILLATOR ENGINE

OHM-1 is a **polyphonic subtractive synthesizer.** Each voice can use up to **8 simultaneous oscillators**, all tuned to the same note but with optional detuning spread. The oscillators are rendered in real time using the Web Audio API at the system's native sample rate (typically 44.1 kHz or 48 kHz).

2.2 WAVEFORMS

Four classic analog waveforms are available. Select the desired waveform using the waveform selector buttons in the SOURCE panel.

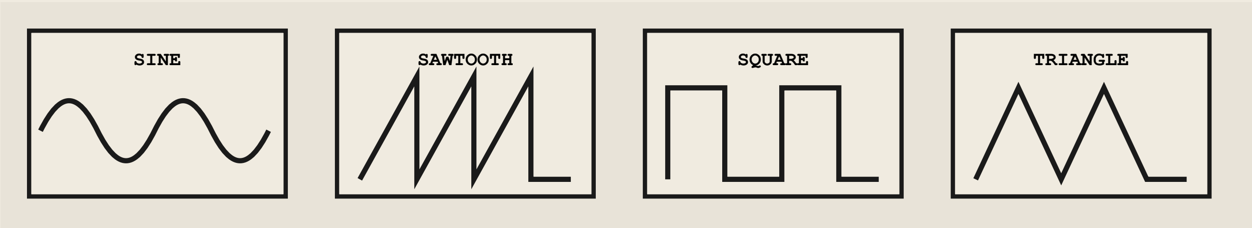

FIG. 2-1 — AVAILABLE OSCILLATOR WAVEFORMS

Waveform	Harmonic Content	Character
Sine	Fundamental only	Pure, clean, flute-like
Sawtooth	All harmonics (odd + even)	Bright, buzzy, string/brass-like
Square	Odd harmonics only	Hollow, nasal, clarinet-like
Triangle	Odd harmonics, lower amplitude	Soft, mellow, between sine and square

2.3 OSCILLATOR COUNT & SPREAD (DETUNE)

OSC COUNT — Set the number of oscillators per voice using the [−] and [+] buttons. Range: 1 to 8. Multiple oscillators create a fuller, thicker sound. Gain is automatically compensated to maintain consistent perceived loudness.

SPREAD — Detunes the oscillators symmetrically around the center pitch. Range: 0–100 cents. At 0%, all oscillators are in unison. Higher values produce a wide, detuned "supersaw" effect. Dynamic gain adjusts for phase coherence.

2.4 NOISE GENERATOR

The **UTILITY** module provides a built-in noise generator, independent of the oscillators. Three noise colors are available via the **COLOR** dropdown. Noise level is set with the **NOISE LVL** knob (0.0 – 1.0). Noise mixes directly into the signal before the filter.

Color	Spectral Density	Character & Use
White	Flat (0 dB/octave)	Bright hiss. Air, hi-hat layering, breath
Pink	−3 dB/octave	Balanced, natural. Wind, ocean, percussion
Brown	−6 dB/octave	Deep, dark rumble. Bass layering, thunder, sub texture

TIP

Mix a small amount of White noise with a Sine wave oscillator and open the filter slightly to create realistic flute and wind instrument sounds. Use Brown noise layered under a bass patch for added sub-frequency weight.

2.5 SAMPLER MODE

Switch the SOURCE module to **SAMPLER** mode to play back audio samples chromatically instead of using oscillators. The sampler uses the same filter and amplifier ADSR pipeline as oscillator mode. Velocity center point is 64 (MIDI), corresponding to unity gain.

3 UTILITY MODULE

3.1 VOICING MODE

POLY — Polyphonic mode (default). Multiple notes can play simultaneously, each with its own independent voice, filter, and envelope. Suitable for chords and pads.

MONO — Monophonic mode. Only one note sounds at a time. When a new note is triggered, the previous voice is released immediately. Ideal for bass lines and monophonic leads.

4 FILTER MODULE

OHM-1 uses a **Lowpass Biquad Filter** per voice. The filter shapes the tonal character of the sound by attenuating frequencies above the cutoff point. Its behavior is modulated by a dedicated ADSR envelope.

4.1 CUTOFF & RESONANCE

CUTOFF	Sets the filter's cutoff frequency. Range: **20 Hz – 10,000 Hz** (logarithmic scale). At low values the sound is dark and muffled; at high values it is bright and open. Default: 2,000 Hz.
RES (Q)	Resonance emphasizes frequencies around the cutoff point. Range: **1.0 – 20.0**. Higher values produce a pronounced peak or self-oscillation at the cutoff frequency. Default: 1.0.
ENV AMT	Controls how much the filter envelope modulates the cutoff frequency. Range: **0 – 100%**. At 100%, the envelope can sweep the filter up to 5,000 Hz above the sustain frequency. Default: 50%.

4.2 FILTER ENVELOPE (ADSR)

The Filter Envelope shapes how the cutoff frequency evolves over time during a note. The curve below shows a typical filter envelope sweep.

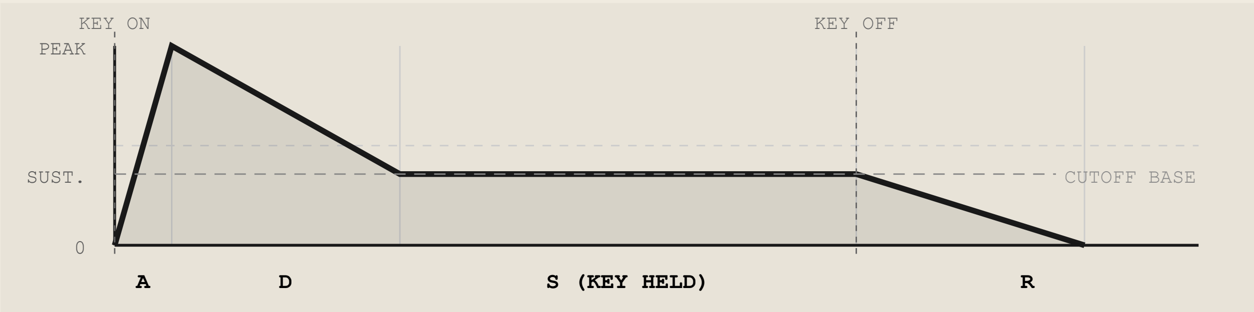

FIG. 4-1 — FILTER ENVELOPE SHAPE

Parameter	Range	Description
ATTACK	0 – 1.0 s	Time to reach peak filter frequency
DECAY	0 – 1.0 s	Time to fall from peak to sustain level
SUSTAIN	0 – 1.0 (0–100%)	Held filter level during key press
RELEASE	0.02 – 1.0 s	Time to fall from sustain to closed (min 20 ms)

5 AMPLIFIER MODULE

The Amplifier Module controls the overall loudness and the volume envelope of each voice. The **Amplitude Envelope** determines how the volume evolves from note-on to note-off.

5.1 MASTER VOLUME

MASTER VOL — Sets the final output level after all processing. Range: **0.0 – 1.0.** Default: 0.5. This parameter is MIDI-mappable and affects the output of all active voices simultaneously.

AUTOMATIC GAIN COMPENSATION

OHM-1 automatically adjusts per-voice gain based on the number of active voices and oscillators to prevent clipping. More oscillators and more voices result in proportionally reduced individual voice levels. Velocity (0–127) applies an additional gain curve with an exponent of 1.2 for expressive dynamic response.

5.2 AMPLITUDE ENVELOPE (ADSR)

The Amplitude Envelope shapes the volume contour of each note. A minimum release time of **5 ms** is enforced to prevent audible clicks during fast releases.

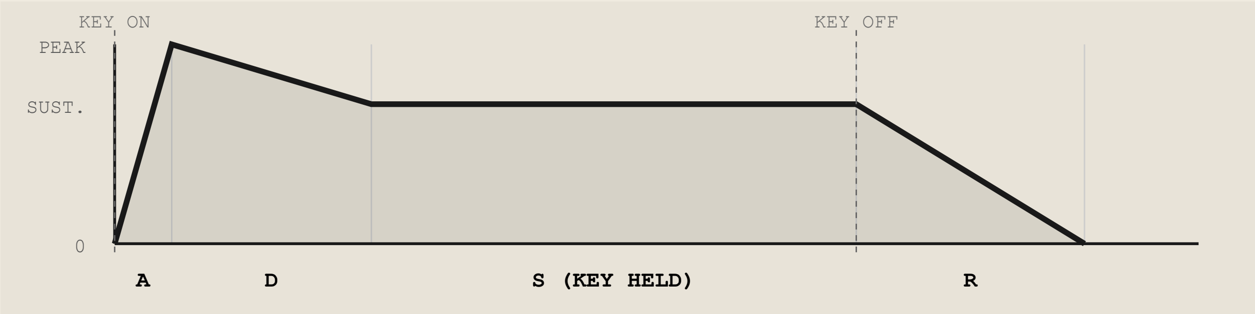

FIG. 5-1 — AMPLITUDE ENVELOPE SHAPE (DEFAULT: A=0.15S, D=0.6S, S=80%, R=5MS)

Parameter	Range	Default	Description
ATTACK	0 – 1.0 s	0.15 s	Time from note-on to maximum volume
DECAY	0 – 1.0 s	0.6 s	Time from peak to sustain level
SUSTAIN	0 – 1.0	0.8 (80%)	Volume level held during key press
RELEASE	0.005 – 1.0 s	0.005 s	Time to fade to silence after key release (min 5 ms)

DESIGNING PATCHES — ADSR GUIDE

PADS:

Long A (0.5–1.0s), moderate D, high S (0.7–1.0), long R (0.5–1.0s).

PLUCKS / LEADS:

Short A (0.0s), moderate D (0.2–0.5s), low S (0.1–0.3), short R.

STRINGS:

Moderate A (0.1–0.3s), long D, high S, moderate R.

BASS:

Short A, short D, moderate-high S, short R.

6 LFO & MODULATION

OHM-1 provides **two independent Low-Frequency Oscillators (LFOs)** for modulating audio parameters over time. LFOs add movement and expression to static sounds — vibrato, tremolo, filter sweeps, and rhythmic effects.

6.1 LFO 1 & LFO 2

RATE	LFO oscillation speed. Range: **0 – 20 Hz**. Slow rates (0.1–2 Hz) for sweeps; fast rates (5–20 Hz) for vibrato/tremolo effects.
DEPTH	Modulation intensity. Range: **0 – 1.0** (normalized). Actual modulation depth depends on the selected destination. At 0, the LFO has no effect.

Waveforms — same four waveforms as the oscillators:

- **Sine** — smooth, natural modulation
- **Triangle** — linear, clean sweep
- **Square** — hard switching between two values
- **Sawtooth** — one-directional ramp

6.2 MODULATION DESTINATIONS

Each LFO can target one destination at a time. Select the destination via the dropdown below the LFO waveform buttons.

Destination	Modulates	Max Depth Scaling	Effect
NONE	—	—	LFO is off / no modulation
FILTER CUTOFF	Filter frequency	±4,000 Hz	Auto-wah, filter sweep, talking synth
PITCH	Oscillator detune	±100 cents	Vibrato, pitch shimmer, chorus-like effect
VOLUME	Master gain	±0.3	Tremolo, pulsating volume
DELAY TIME	Delay effect time	±0.02 s	Tape flutter, subtle warble on delays

6.3 KEY SYNC

When **KEY SYNC** is enabled, the LFO resets its phase to zero each time a new MIDI note is triggered. This ensures the modulation always starts at the same point in its cycle, creating rhythmically consistent effects. When disabled, the LFO runs freely and continuously regardless of note input.

LFO	Default Waveform	Default Rate	Default Key Sync	Default State
LFO 1	Sine	0.0 Hz	Off (free-running)	Enabled
LFO 2	Triangle	2.0 Hz	On (synced)	Disabled

7 STEP SEQUENCER

OHM-1 includes an integrated **Step Sequencer** capable of generating melodic and rhythmic patterns independently of or alongside MIDI input. It triggers note-on/note-off events directly into the voice engine, fully respecting all synthesizer parameters in real time.

7.1 GRID LAYOUT

Columns: 8 steps per row (each step = one 16th note).

Rows: 1 to 8 rows. Each row adds 8 more steps. Click **+ ADD ROW** to extend. Default: 4 rows (32 steps total).

Maximum: 8 rows × 8 columns = **64 steps**.

Each step stores:

- **Active** — whether the step triggers a note
- **Note** — MIDI note number (0–127)
- **Velocity** — MIDI velocity (0–127)

7.2 BPM & TIMING

Set tempo with the **BPM** display (range: 10–999 BPM, default: 90 BPM). Each step = one 16th note. At 90 BPM, each step lasts approximately 166 ms. Note duration is 75% of the step length. Scheduling uses the Web Audio API's sample-accurate clock with a 100 ms look-ahead window for reliable timing.

7.3 PLAYBACK & LOOP BEHAVIOR

Press ▶ **PLAY** to start and ■ **STOP** to halt the sequencer. Playback begins from step 0. After reaching the last active step in a row, the sequencer scans ahead — if no further active steps exist in the remaining rows, it loops immediately back to step 0 rather than advancing through empty rows.

7.4 PROGRAMMING STEPS

Toggle Step	Click any step cell to activate (highlighted) or deactivate it.
Drag & Drop	Drag an active step to another position to move its note and velocity data.
Copy / Paste	**Ctrl+C** to copy a selected step, **Ctrl+V** to paste into the armed step position.
Delete Step	**Backspace** to clear the selected step.
Transpose	**Arrow Up / Down** to shift the selected step's note ±1 semitone.

7.5 MIDI STEP RECORDING

OHM-1 supports live MIDI step recording:

1. Click on a step cell to arm it (highlighted border).
2. Play any note on your MIDI controller.
3. The step captures the MIDI note number and velocity.
4. The armed position automatically advances to the next available step.

7.6 DEFAULT PATTERN

On initialization, Row 1 is pre-loaded with a C major scale ascending: **C4 D4 E4 F4 G4 A4 B4 C5** — all steps active at velocity 127. Rows 2–4 are empty.

8 EFFECTS RACK

The Effects Rack processes the combined output of all active voices in the following fixed order: **Distortion → Chorus → Delay → Reverb → Compressor.** Each effect can be enabled or disabled independently. When disabled, its mix is set to 0 and it is bypassed.

8.1 DISTORTION

DRIVE	Amount of saturation applied. Range: 0 – 1.0. Internally scales to 0–1000 drive gain with a sigmoid waveshaper curve. Automatic makeup gain compensates for volume increase (reduces from 1.0 → 0.1 as drive increases).
TONE	Post-distortion low-pass filter frequency. Range: 500 Hz – 20,000 Hz (exponential). At 1.0 the sound is fully bright; lower values roll off high-frequency distortion artifacts.
MIX	Dry/wet blend. Range: 0 – 1.0. Equal-power crossfade. 0 = clean signal only, 1.0 = fully distorted.

8.2 CHORUS

The Chorus uses a modulated delay line to create a doubling/thickening effect, simulating multiple detuned voices.

RATE	LFO modulation speed. Range: 0.1 – 8.0 Hz. Slow rates produce a gentle shimmer; fast rates create a vibrato-like effect.
DEPTH	Modulation depth of the internal delay time. Range: 0 – 1.0 (internally 0–15 ms offset from 20 ms base delay).
MIX	Dry/wet blend. Default: 0.15 (subtle). Equal-power crossfade.

8.3 DELAY

A feedback delay line with configurable time, feedback amount, and wet level.

TIME	Delay repeat time. Range: 0 – 1.0 seconds. Feedback path carries the delayed signal back for additional repeats.
FEEDBACK	Number of repeats / tail length. Range: 0 – 0.9 (capped at 90% to prevent runaway feedback). Higher values create long, decaying tails.
MIX	Dry/wet blend. Range: 0 – 1.0. Default: 1.0 when enabled.

8.4 REVERB

A convolution reverb using a synthetically generated impulse response. Provides a sense of acoustic space from small rooms to large halls.

MIX	Dry/wet blend. Range: 0 – 1.0. Default: 0.5. Equal-power crossfade.
DECAY	Reverb tail length. Range: 0.1 – 5.0 seconds. The impulse response is regenerated when this value changes (100 ms debounce). Short values simulate small rooms; long values simulate halls or chambers.

EFFECTS ORDER NOTE

The effects chain order is fixed: Distortion is applied first (pre-reverb), ensuring that distorted signal — rather than the reverb tail — is processed. This is the traditional analog insert chain order. To achieve a "reverb before distortion" effect, consider reducing the reverb mix on the reverb unit and using the distortion's tone control to shape the combined result.

9 MIDI IMPLEMENTATION

OHM-1 uses the **Web MIDI API** to receive MIDI data from connected hardware controllers. Select your MIDI device from the **MIDI IN** dropdown in the top navigation bar. Multiple devices are supported simultaneously (shown as "+N" when more than one is active).

9.1 MIDI INPUT & NOTE HANDLING

Message	Status Byte	Description
Note On	0x90 + ch, vel > 0	Triggers a new voice. Each note spawns an independent voice. Velocity (0–127) scales the voice gain with an exponent of 1.2.
Note Off	0x80 + ch, or 0x90 vel=0	Releases the corresponding voice. The amplitude envelope enters its Release phase. After the release time, the voice is deallocated.
Control Change	0xB0 + ch	Routes CC values (0–127, normalized to 0.0–1.0) to mapped synthesizer parameters.
MIDI Clock	0xF8	Ignored. OHM-1 uses its own internal BPM clock.

9.2 MIDI LEARN — CC MAPPING

Any MIDI CC control from your hardware can be mapped to a synthesizer parameter using the **MIDI Learn** function.

To map a CC to a parameter:

1. Click **MIDI MAPPINGS** in the top bar to open the mapping panel.
2. Click the **LEARN** button next to the desired parameter.
3. The button turns active (lit) — OHM-1 is now listening for the next CC message.
4. Move any knob, fader, or slider on your MIDI controller.
5. The CC number is captured and the mapping is confirmed automatically.
6. To remove a mapping, click the **UNMAP** button next to the parameter.

MIDI MAP VIEW

Click
MIDI MAP

in the top bar to view the complete current mapping table — all parameters and their assigned CC numbers at a glance. Mappings persist as long as the browser session is active.

9.3 MAPPABLE PARAMETERS

Category	Parameter	CC Range → Internal Range
Amp Envelope	ATTACK	0–127 → 0–1.0 s
	DECAY	0–127 → 0–1.0 s
	SUSTAIN	0–127 → 0–1.0
	RELEASE	0–127 → 0–1.0 s
Filter Envelope	FILTER ATTACK	0–127 → 0–1.0 s
	FILTER DECAY	0–127 → 0–1.0 s
	FILTER SUSTAIN	0–127 → 0–1.0
	FILTER RELEASE	0–127 → 0–1.0 s
Filter	FILTER FREQUENCY	0–127 → 20–10,000 Hz (log)
	FILTER RESONANCE	0–127 → 0–20
	FILTER ENV AMOUNT	0–127 → 0–100%
	MASTER GAIN	0–127 → 0–1.0
Oscillators	OSC COUNT	0–127 → 1–8
	SPREAD (DETUNE)	0–127 → 0–100 cents
LFO 1	LFO1 RATE	0–127 → 0–20 Hz
	LFO1 DEPTH	0–127 → 0–1.0
LFO 2	LFO2 RATE	0–127 → 0–20 Hz
	LFO2 DEPTH	0–127 → 0–1.0
Distortion	DRIVE	0–127 → 0–1.0
	TONE	0–127 → 0–1.0
	MIX	0–127 → 0–1.0
Reverb	MIX	0–127 → 0–1.0
	DECAY	0–127 → 0.1–5.0 s
Delay	TIME	0–127 → 0–1.0 s
	FEEDBACK	0–127 → 0–0.9
	MIX	0–127 → 0–1.0
Chorus	RATE	0–127 → 0.1–8.0 Hz
	DEPTH	0–127 → 0–1.0
	MIX	0–127 → 0–1.0

APPENDIX A PARAMETER REFERENCE

Module	Parameter	Min	Max	Default	Unit	MIDI
Source	Waveform	Sine / Sawtooth / Square / Triangle			—	No
Source	OSC Count	1	8	2	integer	Yes
Source	Spread	0	100	0	cents	Yes
Utility	Noise Level	0.0	1.0	0.0	linear	No
Utility	Noise Color	White / Pink / Brown			—	No
Utility	Voicing	Poly / Mono			—	No
Filter	Cutoff	20	10,000	2,000	Hz (log)	Yes
Filter	Resonance	1.0	20.0	1.0	Q	Yes
Filter	Env Amount	0	1.0	0.5	%	Yes
Filter Env	Attack	0	1.0	0.3	s	Yes
Filter Env	Decay	0	1.0	0.0	s	Yes
Filter Env	Sustain	0	1.0	0.05	linear	Yes
Filter Env	Release	0.02	1.0	0.02	s	Yes
Amplifier	Master Vol	0	1.0	0.5	linear	Yes
Amp Env	Attack	0	1.0	0.15	s	Yes
Amp Env	Decay	0	1.0	0.6	s	Yes
Amp Env	Sustain	0	1.0	0.8	linear	Yes
Amp Env	Release	0.005	1.0	0.005	s	Yes
LFO 1	Rate	0	20	0.0	Hz	Yes
LFO 1	Depth	0	1.0	0.0	linear	Yes
LFO 2	Rate	0	20	2.0	Hz	Yes
LFO 2	Depth	0	1.0	0.0	linear	Yes
Sequencer	BPM	10	999	90	BPM	No
Sequencer	Rows	1	8	4	integer	No
Distortion	Drive	0	1.0	0.5	linear	Yes
Distortion	Tone	500	20,000	20,000	Hz (exp)	Yes
Distortion	Mix	0	1.0	1.0	linear	Yes
Chorus	Rate	0.1	8.0	0.27	Hz	Yes
Chorus	Depth	0	1.0	0.2	linear	Yes
Chorus	Mix	0	1.0	0.15	linear	Yes
Delay	Time	0	1.0	0.5	s	Yes
Delay	Feedback	0	0.9	0.3	linear	Yes
Delay	Mix	0	1.0	1.0	linear	Yes
Reverb	Mix	0	1.0	0.5	linear	Yes
Reverb	Decay	0.1	5.0	2.0	s	Yes